

Single-Camera Bird’s Eye View Scene Reconstruction with Object Detection

Irene Williams

Abstract

Bird’s Eye View (BEV) representations are widely used in applications such as autonomous driving, where understanding spatial relationships between objects is critical. Most existing BEV systems rely on multiple calibrated cameras and often incorporate depth estimation or LiDAR data. In this project, we explore a simplified pipeline that generates a BEV-style representation using only a single uncalibrated camera capturing images while rotating in place. The system constructs a panorama using both OpenCV stitching and a custom cylindrical stitching pipeline. Object detection is performed using a YOLOv8 model, and detections are mapped into panorama coordinates using homographies. Duplicate detections are removed, and a BEV map is generated by applying a perspective transform to the ground region of the panorama. The results demonstrate that a visually consistent BEV representation can be produced without depth estimation or multi-camera calibration, although geometric accuracy remains approximate.

1. Introduction

Bird’s Eye View (BEV) representations are commonly used in computer vision tasks that require spatial reasoning, especially in autonomous driving systems. These systems typically rely on multiple cameras and depth information to accurately reconstruct the environment. For example, modern autonomous driving pipelines often fuse camera images with LiDAR data to generate accurate 3D scene representations [2]. Additionally, many recent BEV approaches rely on learned depth estimation or multi-view geometry to project image features into a top-down space [4].

While these approaches are effective, they require calibrated sensors and complex system setups. In contrast, this project investigates whether a simpler system can approximate a BEV representation using only a single camera.

The idea is to capture a sequence of images while rotating in place, stitch them into a panorama, detect objects in the scene, and then map those detections into a top-down representation. This removes the need for depth estimation or multiple viewpoints but introduces several challenges.

The main challenges in this project include stitching im-

ages without known camera parameters, handling drift, distortion, and misalignment in panoramas, mapping detections consistently across multiple images, and constructing a BEV representation without true depth information.

To address these challenges, we implement a full pipeline that combines classical computer vision techniques with modern object detection. First, overlapping images are stitched into a panorama using both OpenCV’s built-in stitching algorithm and a custom cylindrical stitching method, with additional corrections for drift, distortion, and orientation. Object detection is then performed on the original images using a YOLOv8 model, and detections are mapped into the panorama using homographies. Duplicate detections across overlapping views are removed using a distance-based clustering approach. Finally, a BEV representation is generated by applying a perspective transform to the ground region of the panorama, and detected objects are projected into this top-down view. This approach avoids the need for calibration or depth estimation while still producing a consistent spatial representation of the scene.

2. Related Work

Image stitching has been extensively studied in computer vision, particularly for constructing panoramic images from overlapping views. Classical approaches rely on detecting features in each image, matching them across pairs, and estimating geometric transformations such as homographies to align the images [6]. These methods typically assume either planar scenes or purely rotational camera motion, which simplifies the alignment problem.

More advanced stitching methods, such as the work by Brown and Lowe [1], use invariant local features (e.g., SIFT) to automatically match images without requiring a fixed ordering. This allows stitching to work even when images are unordered or captured with varying orientations. These methods also introduce techniques such as gain compensation and automatic straightening, which help improve visual quality.

In addition to feature-based approaches, several surveys have highlighted the main challenges in image stitching, including parallax, exposure differences, and misalignment in overlapping regions [3]. These issues often lead to artifacts such as ghosting or visible seams, which must be addressed

through blending and geometric correction. In this project, many of these challenges appear in practice, especially due to the lack of camera calibration and imperfect image capture.

Cylindrical projection is a common technique used in panorama construction, particularly for images captured with rotational motion. By projecting images onto a cylindrical surface, horizontal distortion is reduced and alignment becomes more stable [6]. This directly motivates the custom cylindrical stitching pipeline used in this project, where images are warped before alignment.

Object detection has also seen major advancements with deep learning models such as YOLO [5]. These models are widely used due to their ability to perform fast and accurate detection in real-time. In this project, YOLOv8 is used to detect objects independently in each image before mapping them into the panorama.

Finally, BEV perception has been explored heavily in the context of autonomous driving. Recent methods such as Lift-Splat-Shoot [4] project image features into a BEV representation using learned depth estimation. Similarly, multi-sensor approaches such as BEVFusion combine camera and LiDAR data to produce accurate spatial representations [2]. These methods achieve high accuracy but rely on multiple sensors and calibration.

In contrast, this project focuses on a much simpler setup. Instead of using depth estimation or multiple cameras, it relies entirely on geometric transformations derived from image stitching and homographies. While this leads to less accurate spatial reconstruction, it demonstrates that a reasonable BEV-style representation can still be achieved using only a single rotating camera.

3. Approach

The pipeline consists of three main stages: panorama construction, object detection and mapping, and BEV generation. The system is implemented in Python using OpenCV for image processing and the Ultralytics implementation of YOLOv8 for object detection.

3.1. Panorama Construction

The first goal is to combine all input images into a single panorama. We implemented two approaches: OpenCV stitching (primary) and a custom cylindrical stitching method (fallback).

In the cylindrical method, each image is warped onto a cylindrical surface using an estimated focal length. This reduces distortion caused by rotational motion. Adjacent images are aligned using SIFT feature matching, and relative translations are estimated using RANSAC. These translations are accumulated across the sequence to place all images in a shared coordinate system.

One major issue with this approach is that small alignment errors accumulate, causing the panorama to tilt. To fix this, we apply a linear vertical correction to all images so that the panorama remains level.

Blending is performed using a distance-based weighting scheme rather than simple averaging. Pixels near the center of each image are weighted more heavily than pixels near the edges, which reduces blurriness in overlapping regions.

For blending adjacent images, we use a distance-based weighting scheme. Pixels near the center of each image are weighted more heavily than pixels near the edges, which reduces blurriness in overlapping regions.

After stitching, several cleanup steps are applied:

- cropping to remove black borders,
- trimming repeated regions caused by wraparound,
- perspective correction to remove slant,
- and rotation using ground-line detection to level the panorama.

Figure 1 shows the cylindrical panorama before cleanup, while Figure 2 shows the final result.



Figure 1. Cylindrical panorama before cleanup (drift, overlap, and distortion visible).



Figure 2. Final cylindrical panorama after drift correction, trimming, and leveling.

In the final pipeline, OpenCV stitching is attempted first on both original and preprocessed images. If both fail an aspect ratio test, the cylindrical result is used.

3.2. Object Detection and Mapping

Object detection is performed on the original images using YOLOv8. Running detection on the original images avoids distortion issues that occur in panoramas. Figure 3 shows an example detection result on a single image.

To understand how images contribute to the panorama, Figure 4 shows the approximate placement of individual images within the stitched result.

Each detection is mapped into the panorama using homographies. For each image, ORB feature matching is used to align the image with the panorama, and a homography H



Figure 3. Example YOLOv8 detection on an input image.



Figure 4. Approximate placement of individual images within the panorama.

is estimated. The center of each bounding box is projected using:

$$p' = Hp$$

Since objects appear in multiple overlapping images, duplicate detections must be removed. Detections of the same class are grouped using a circular distance threshold:

$$d = \sqrt{dx^2 + dy^2}$$

Horizontal wraparound is handled by computing distances modulo the panorama width. The highest-confidence detection in each group is kept. Figure 5 shows detections after mapping into the panorama.



Figure 5. Detections mapped into panorama coordinates after deduplication.

3.3. BEV Generation

The BEV map is constructed directly from the panorama. Instead of estimating depth, we assume that the lower portion of the panorama corresponds to the ground.

A vertical band from the lower part of the panorama is selected and mapped into a circular BEV canvas. Each pixel in the BEV image samples from the panorama based

on its angle and distance from the center. This produces a surround-view style background. Figure 6 shows the BEV map before adding detections.



Figure 6. BEV background generated from the panorama without detections.

After mapping detections into panorama coordinates, they are projected into the BEV map using the same angular mapping, where the horizontal position determines angle and the vertical position determines approximate distance. This approach does not produce metric distances, but it creates a consistent and visually meaningful top-down representation of the scene.

4. Results

The system was tested on 5 scenes, each consisting of approximately 8–10 landscape images captured while rotating in place. Results are evaluated qualitatively by examining panorama quality, detection consistency, and the final BEV representation.

4.1. Panorama Results

OpenCV stitching produced the highest-quality panoramas when it succeeded, particularly in scenes with strong feature overlap. However, it occasionally failed to include all images or produced incomplete results.

The cylindrical method provided a reliable fallback. While the initial cylindrical panorama contained drift, overlap, and distortion (Figure 1), the additional corrections significantly improved the final result (Figure 2). The final panoramas can be viewed in Figure 7



Figure 7. Final panoramas.

4.2. Detection Mapping Results

Mapping detections into the panorama produced a consistent representation of object locations across multiple views. The use of homographies allowed detections from different images to align into a shared coordinate system.

However, some errors remained:

- small misalignments due to imperfect homographies,
- duplicate detections when objects appeared across multiple images,
- occasional missed detections near image boundaries.

These errors can be viewed in Figure 8.

4.3. BEV Results

The final BEV maps (shown in Figure 9) provide a clear top-down visualization of the scene.



Figure 8. Misalignments, duplicate detections, and missed detections in panorama.

Objects are placed at approximately correct angular positions relative to the camera, and the background generated from the panorama provides useful context.

However, the BEV representation is not metrically accurate. Since no depth estimation is used, distances are approximated based on image position, which leads to compression of far objects and some inconsistency in scale. With that being said, the system seems to produce consistent and interpretable BEV visualizations across multiple scenes.

The dataset, intermediate results, final results, and code-base are available at: [Github Link](#)



Figure 9. Final BEV map with projected detections.

5. Discussion and Conclusions

This project explored whether a Bird’s Eye View (BEV) representation could be generated using only a single rotating camera, without calibration or depth estimation. Overall, the results show that it is possible to produce a consistent and visually meaningful top-down representation using a combination of panorama stitching, object detection, and geometric transformations.

One of the main insights from this project is that panorama construction is the most critical step in the pipeline. When the panorama is well-aligned and complete, the later stages (mapping detections and generating the BEV) work significantly better. OpenCV stitching generally produced the best results, but it was not always reliable. The custom cylindrical method, while less accurate, was important as a fallback and became much more effective after adding drift correction, trimming, and leveling.

Another important observation is that mapping detections using homographies can work well in certain situations, but it does have limitations. In scenes with fewer objects, especially when the objects are visually separated and do not overlap much across views, the mapping step tends to be more reliable. However, when objects are tightly packed, partially occluded, or repeated across multiple overlapping images, the process becomes much harder. In those cases, even small homography errors can move the projected detection center enough to create duplicate detections or cause noticeable placement errors.

By projecting detections into the panorama, it becomes

possible to combine information from multiple views into a single coordinate system. However, small errors in homography estimation can still lead to slight misalignment in object positions. As a result, this method works best as an approximate way to unify detections across views rather than as a perfectly precise localization method.

The BEV generation step demonstrates that a reasonable top-down representation can be created using only image-based transformations. The use of a panorama-derived background significantly improves the realism of the BEV map compared to a blank canvas. However, because no depth estimation is used, object distances are only approximate. This results in compression of distant objects and some inconsistency in scale. Further, due to the way the images were captured, the inner radius of the BEV map is relatively large. Since the camera was held near eye level, the ground directly near the camera was not captured in the images, which creates a gap in the center of the BEV representation.

There are several limitations to the current approach:

- The accuracy of the system depends heavily on the quality of the panorama. Errors in stitching directly affect detection mapping and BEV projection.
- The lack of depth information means that object placement in the BEV map is not metrically accurate.
- The method assumes a roughly planar ground surface, which may not hold in more complex environments.
- The system is sensitive to how the images are captured. If the camera motion is not close to pure rotation, additional distortion is introduced.
- The images are expected to be taken in a consistent order with sufficient overlap. If the sequence is out of order or the overlap between images is too small, stitching and detection mapping become less reliable.
- Homography estimation can introduce small errors, especially in scenes with repetitive textures or limited feature matches.
- Detection quality also affects the final result, since missed or incorrect detections propagate into the BEV map.

Despite these limitations, the system demonstrates that meaningful spatial structure can be recovered from a simple image sequence. This suggests that even without complex sensor setups, it is possible to approximate BEV-style representations using relatively simple techniques.

Future work could improve the system in several ways. One natural extension would be to incorporate depth estimation, either through monocular depth models or multi-

view geometry, which would allow for more accurate object placement in the BEV map. Another improvement would be to use global optimization for panorama alignment, rather than relying on pairwise matching and accumulated offsets, which could reduce drift and improve consistency.

Additionally, using more advanced stitching or blending techniques could help reduce visual artifacts in the panorama. The BEV generation step could also be improved by learning a mapping from image space to BEV space, rather than relying on a fixed geometric transform. Finally, improving the data collection process, such as capturing images at a lower height or ensuring more consistent rotation, could help reduce some of the geometric limitations observed in the current results.

6. Statement of Individual Contribution

This project was completed individually. I was responsible for all aspects of the work, including data collection, implementation of the panorama stitching methods, object detection and mapping, BEV generation, debugging, and evaluation of results. I also handled visualization, experimentation across multiple scenes, and the final write-up.

7. External Resources Used

The following external resources, libraries, and tools were used in this project:

- **OpenCV**

<https://opencv.org>

Used for core image processing operations, including image loading, resizing, cylindrical warping, feature detection (SIFT and ORB), feature matching, homography estimation, edge detection, Hough line detection, and geometric transformations. OpenCV was also used for blending, warping, and constructing panoramas.

- **OpenCV Stitcher Module**

https://docs.opencv.org/4.x/d8/d19/tutorial_stitcher.html

Used as the primary method for panorama stitching. This module performs feature detection, matching, homography estimation, and blending automatically. It was used on both original and preprocessed images to generate high-quality panoramas.

- **Ultralytics YOLOv8**

<https://docs.ultralytics.com>

Used for object detection. A pretrained YOLOv8 model was used to detect objects such as cars, people, and bicycles in each input image. The model outputs bounding boxes, class labels, and confidence scores,

which are later mapped into the panorama and BEV representations.

- **NumPy**

<https://numpy.org>

Used for numerical operations throughout the project, including array manipulation, coordinate transformations, distance calculations, and weighted averaging for blending and deduplication.

- **Matplotlib**

<https://matplotlib.org>

Used for visualizing intermediate and final results, including displaying panoramas, detection outputs, and BEV maps during development.

- **Python Standard Libraries (os, math, glob)**

<https://docs.python.org/3/library>

Used for file handling, directory management, mathematical computations, and loading image datasets from disk.

- **AI Assistance (ChatGPT)**

<https://openai.com/chatgpt>

Used as a development aid for debugging, refining implementations, and generating example code for specific components such as panorama stitching, homography mapping, and BEV visualization. All final code was integrated and modified to fit the specific requirements of the project.

References

- [1] Matthew Brown and David G Lowe. Automatic panoramic image stitching using invariant features. *International Journal of Computer Vision*, 74(1):59–73, 2007. 1
- [2] Zhijian Liu, Haotian Tang, Alexander Amini, Xinyu Yang, Huizi Mao, Daniela Rus, and Song Han. Bevfusion: Multi-task multi-sensor fusion with unified bird’s-eye view representation, 2024. 1, 2
- [3] Wei LYU, Zhong ZHOU, Lang CHEN, and Yi ZHOU. A survey on image and video stitching. *Virtual Reality Intelligent Hardware*, 1(1):55–83, 2019. 1
- [4] Jonah Philion and Sanja Fidler. Lift, splat, shoot: Encoding images from arbitrary camera rigs by implicitly unprojecting to 3d. In *ECCV*, 2020. 1, 2
- [5] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection, 2016. 2
- [6] Richard Szeliski. Image alignment and stitching: a tutorial. *Found. Trends. Comput. Graph. Vis.*, 2(1):1–104, Jan. 2006. 1, 2